

An integer-matrix separation proving Conjecture 00000006330

gaochengzhecpu

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The source asks for two matrices with identical determinant and permanent but different integer types, with the separation described through unimodular row transformations. The following pair works even if both unimodular row and column operations are allowed:

$$A = \begin{pmatrix} 1 & 0 \\ 0 & 4 \end{pmatrix}, \quad B = \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix}.$$

They satisfy

$$\det A = \det B = 4, \quad \text{per } A = \text{per } B = 4.$$

Both matrices are already in Smith normal form: their diagonal entries are positive and the first divides the second. Their invariant-factor lists are respectively $(1, 4)$ and $(2, 2)$.

A direct argument establishes the distinction without invoking the Smith normal form classification theorem. For arbitrary integer matrices U, V , every entry of $UBV = 2UV$ is even. Therefore $UBV \neq A$, whose top-left entry is one. In particular, there are no unimodular integer U, V with $UBV = A$. Thus the matrices have distinct integer-equivalence types.

The row-only interpretation is also covered in both orientations. An equation $UB = A$ is impossible by parity. An equation $UA = B$ would require $4U_{22} = 2$ by comparing the bottom-right entries, which is impossible over the integers. Hence neither matrix is obtained from the other by unimodular row transformations. This is the precise separation requested in the source, with a stronger two-sided obstruction as well.

Formal verification. The self-contained Lean 4.19.0 project defines actual integer matrices as functions $\text{Fin } 2 \rightarrow \text{Fin } 2 \rightarrow \mathbb{Z}$. Matrix multiplication, determinant, permanent, and unimodularity are defined explicitly. Row equivalence and two-sided integer equivalence quantify over *all* integer matrices U, V ; these quantifiers are not replaced by a finite search or a bound on their entries.

The key invariant is the predicate that every matrix entry is an integer multiple of two. The file proves that multiplication on either side by any integer matrix preserves this predicate. It then verifies both determinants, both permanents, the actual Smith-diagonal conditions, and the inequivalences. The final existence theorem is `conjecture_6330` in `lean/Main.lean`. It uses only bundled `Std`, with no `sorry`, `admit`, `native_decide`, or added axiom. The printed axiom list is limited to Lean's standard `propext` and `Quot.sound`.